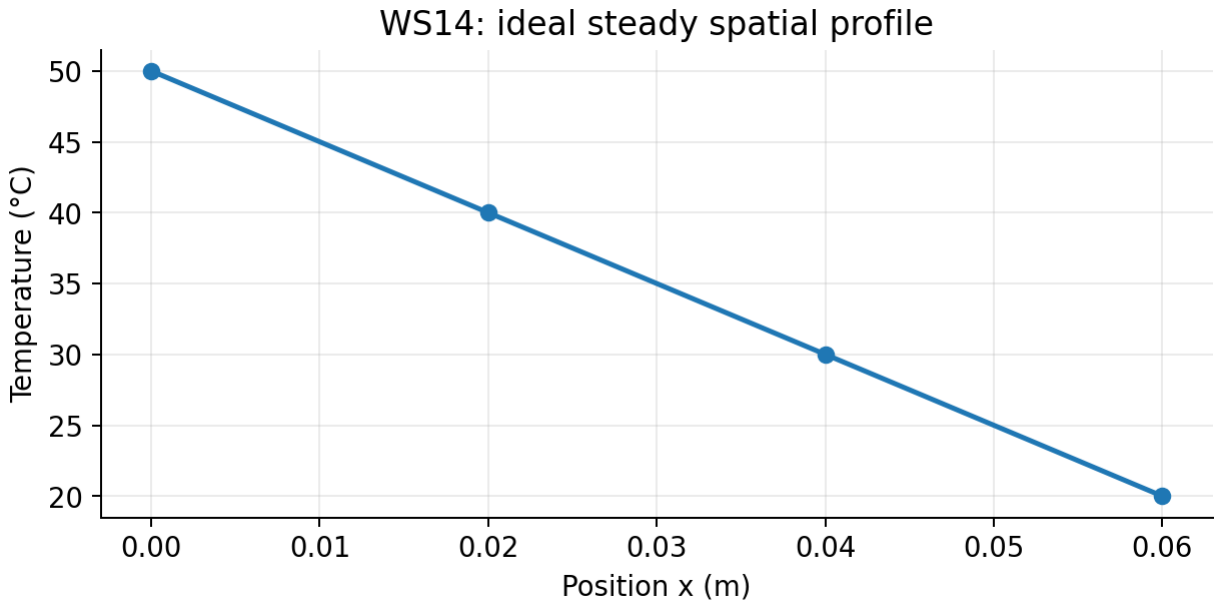
WS14 - Reference page

Keep this page beside the four question pages that follow.

Ideal, uniform, steady one-dimensional profile. Position is measured from the warmer face. These are model values, not time readings.



position_m	temperature_C
0	50
0.02	40
0.04	30
0.06	20

WS14 - Conduction

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 1 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS14 reference page preceding these question pages.

Objectives: Explain conduction microscopically; identify its direction; interpret a temperature profile; calculate a gradient and energy from a constant rate.

WS14-Q01 (2 points) A solid rod warms at one end. Explain how energy can reach the other end without the atoms flowing along the rod.

WS14-Q02 (2 points) A rod has faces at 45 °C and 25 °C. In which direction is net conduction? What happens to the net conducting transfer if the entire rod reaches 25 °C?

WS14 - Conduction

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 2 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS14 reference page preceding these question pages.

WS14-Q03 (2 points) Which statement is correct? A. Only solids conduct. B. Still fluids can conduct. C. Conduction requires bulk circulation. Explain.

WS14-Q04 (2 points) A constant conducting rate is 4.0 W. State what this means in joules and seconds, then calculate the energy transferred in 30 s.

WS14 - Conduction

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 3 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS14 reference page preceding these question pages.

WS14-Q05 (3 points) The WS 14 reference profile is steady. Calculate its gradient magnitude in K/m and identify the direction of net conduction.

WS14-Q06 (3 points) A conducting path transfers 2.5 W for 2.0 min and then 1.5 W for 1.0 min. Calculate total energy.

WS14 - Conduction

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 4 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS14 reference page preceding these question pages.

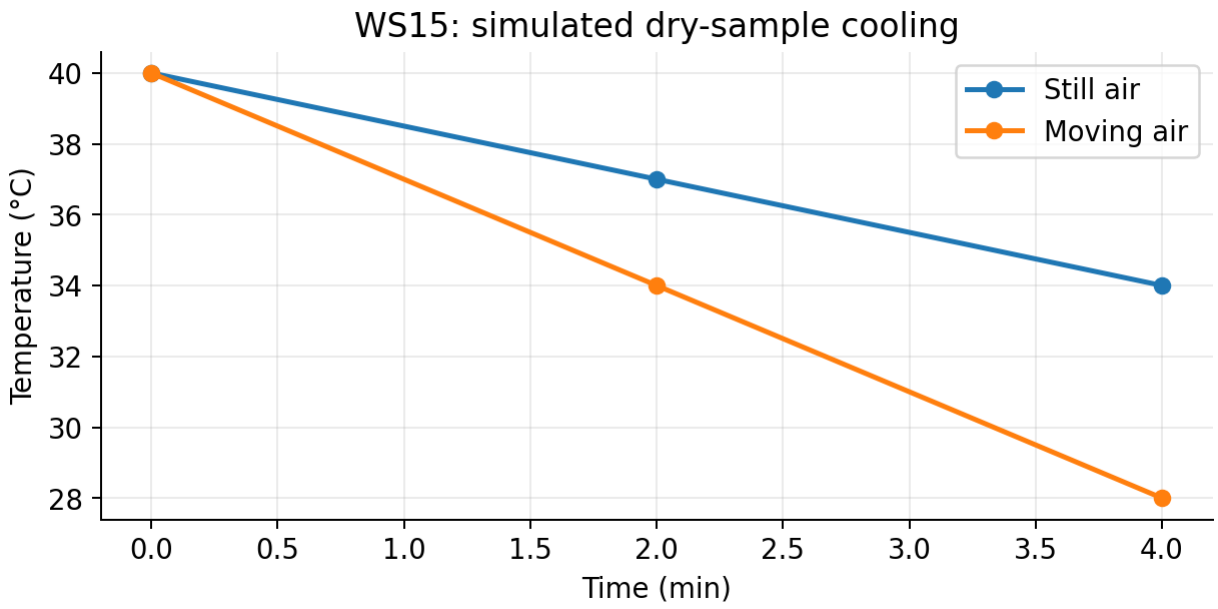
WS14-Q07 (3 points) Two dry room-temperature blocks, metal and wood, have the same measured temperature. The metal feels colder. Explain and name a reason touch is a poor thermometer.

WS14-Q08 (3 points) Challenge: A student compares equal-area slabs of the same material, one 5 mm thick and one 10 mm thick, with identical fixed face-temperature differences. Predict the ideal rate ratio, then explain why a real contact gap could spoil this comparison.

WS15 - Reference page

Keep this page beside the four question pages that follow.

Simulated equal-capacity dry samples, identical starts and room conditions. Only imposed airflow differs in this ideal comparison.



time_min	still_air_C	moving_air_C
0	40	40
2	37	34
4	34	28

WS15 - Convection

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 1 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS15 reference page preceding these question pages.

Objectives: Explain buoyancy-driven circulation and forced circulation; distinguish flow from molecular conduction; analyze rates, density and cooling data.

WS15-Q01 (2 points) Define a fluid and explain why a rigid solid cannot transfer energy by bulk convection within itself.

WS15-Q02 (2 points) Ordinary air well above freezing warms and expands at nearly constant pressure with no mass change. Explain the density change and its possible buoyant motion.

WS15 - Convection

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 2 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS15 reference page preceding these question pages.

WS15-Q03 (2 points) Classify (a) a pump circulating radiator water and (b) air rising above a warm radiator without a fan.

WS15-Q04 (2 points) A sealed flexible model parcel contains 2.0 g of air in 2.5 L. Calculate density.

WS15 - Convection

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 3 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS15 reference page preceding these question pages.

WS15-Q05 (3 points) Sketch a plausible natural-convection loop for water heated gently from below and cooled near the top. Label warmer/less-dense and cooler/denser regions. Assume water stays above 10 °C.

WS15-Q06 (3 points) Use the WS 15 cooling data. Calculate the average 0–4 min slopes for still air and moving air. Which has larger cooling magnitude?

WS15 - Convection

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 4 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS15 reference page preceding these question pages.

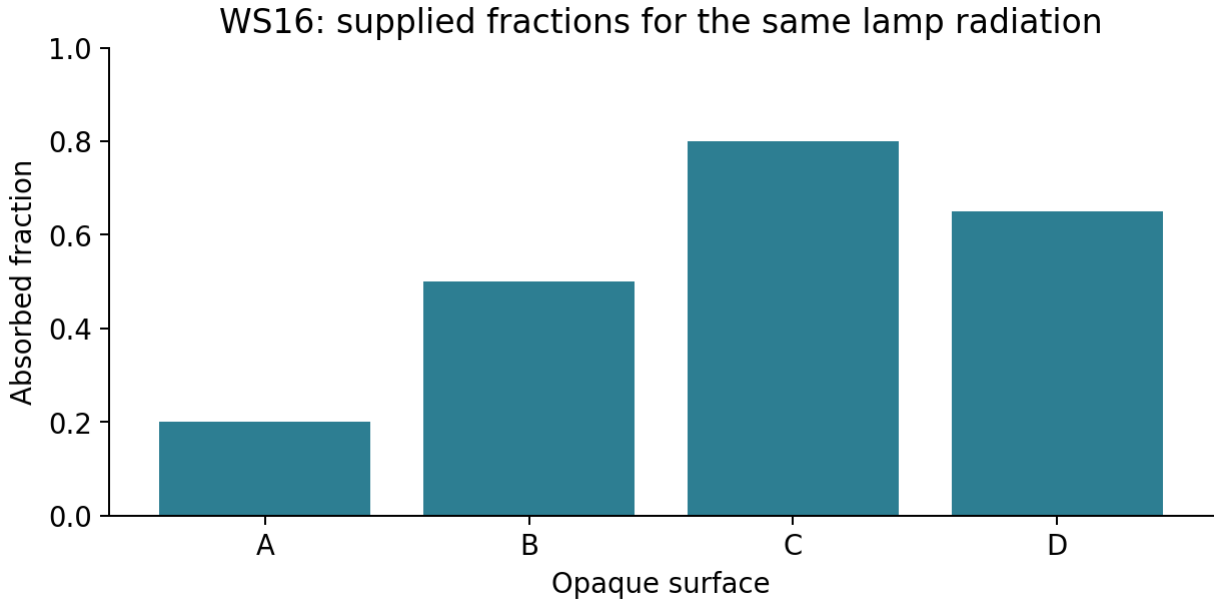
WS15-Q07 (3 points) A fan moves 45 °C dry air over a dry object at 25 °C. Predict the direction of convective transfer and explain why “fans always cool” fails.

WS15-Q08 (3 points) Challenge: Two equal water samples begin at 30 °C. One is heated from below and one from above. Explain which arrangement more readily produces natural overturning. Name two controls needed to compare it fairly.

WS16 - Reference page

Keep this page beside the four question pages that follow.

Supplied absorption fractions for opaque surfaces under the same lamp radiation. These are illustrative properties, not measured claims about named materials.



surface	absorbed_fraction
A	0.2
B	0.5
C	0.8
D	0.65

WS16 – Thermal Radiation

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 1 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS16 reference page preceding these question pages.

Objectives: Explain thermal radiation and net exchange; distinguish absorption, reflection and transmission; calculate energy budgets from supplied incident power and fractions.

WS16-Q01 (2 points) Why can radiation cross a vacuum while convection cannot?

WS16-Q02 (2 points) An ice cube at −5 °C is in a 20 °C room. Does it emit thermal radiation? Explain without using an advanced radiation formula.

WS16 – Thermal Radiation

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 2 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS16 reference page preceding these question pages.

WS16-Q03 (2 points) An opaque surface absorbs 0.30 of a specified incident beam. Find its reflected fraction, assuming no other destination for that beam.

WS16-Q04 (2 points) Two coatings look equally black in visible light. Must they have identical infrared absorption? Explain.

WS16 - Thermal Radiation

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 3 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS16 reference page preceding these question pages.

WS16-Q05 (3 points) A surface receives 20 W of radiation with $\alpha=.60$, $\rho=.25$ and $\tau=.15$. Calculate all three power destinations and check the sum.

WS16-Q06 (3 points) A sample absorbs 9.0 W and emits 4.0 W for 3.0 min; other transfers are negligible and rates are constant. Calculate net radiative heat input.

WS16 - Thermal Radiation

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 4 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS16 reference page preceding these question pages.

WS16-Q07 (3 points) The WS 16 bar chart gives absorbed fractions for four opaque surfaces under the same lamp. Which absorbs the most of a 10 W incident beam, and how much does that surface reflect?

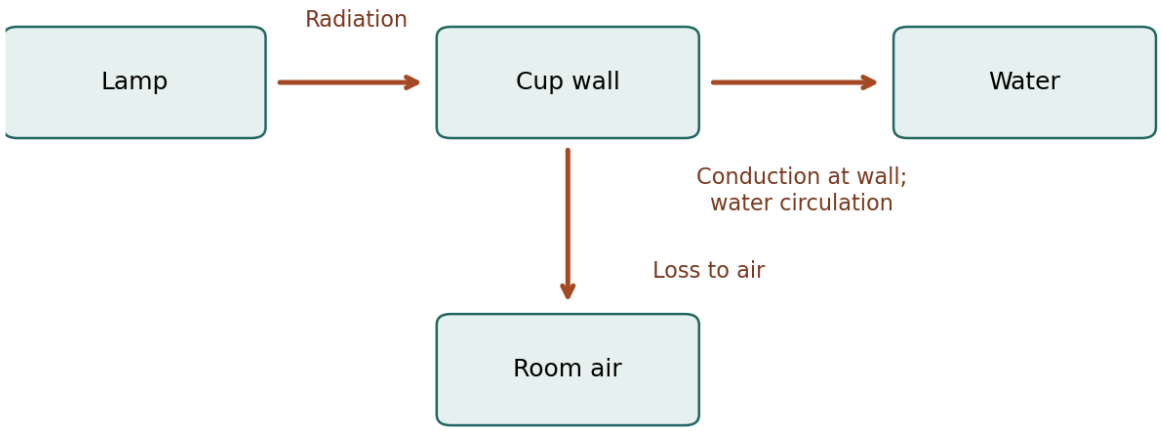
WS16-Q08 (3 points) Challenge: A student uses a 100 W bulb rating and a surface absorptivity of .80 to claim water receives 80 W net. Identify two missing links and state what can actually be calculated if only 15 W reaches that surface.

WS17 - Reference page

Keep this page beside the four question pages that follow.

The selected boundary encloses both cup wall and water; lamp and room air are outside. The diagram shows a simplified set of paths, not every possible transfer.

Sequential input path and separate loss path



path	description
lamp to wall	radiation
wall to water	transfer at wall and water circulation
cup to air	loss to cooler room air

WS17 - Comparing Heat Transfer Mechanisms

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 1 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS17 reference page preceding these question pages.

Objectives: Identify multiple transfer paths; trace energy across chosen boundaries; avoid double counting; evaluate explanations and diagrams.

WS17-Q01 (2 points) Match three stages: (a) energy across empty space from lamp, (b) through a stationary solid cup wall, (c) carried by circulating water.

WS17-Q02 (2 points) A warm cup transfers energy to cooler air while absorbing lamp radiation. Can it warm overall? Explain.

WS17 - Comparing Heat Transfer Mechanisms

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 2 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS17 reference page preceding these question pages.

WS17-Q03 (2 points) Choose the most precise claim: A. Only conduction occurs in a spoon-in-soup setup. B. Conduction carries energy along the solid spoon, while other paths may coexist. Explain.

WS17-Q04 (2 points) In an ideal lossless sequence, 400 J is absorbed by a plate, then passes through a wall, then enters water. How much reaches water, and why not 1200 J?

WS17 - Comparing Heat Transfer Mechanisms

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 3 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS17 reference page preceding these question pages.

WS17-Q05 (3 points) A selected system gains 200 J by one boundary path and 350 J independently by another, while losing 150 J. Find net input and explain why these quantities may be combined.

WS17-Q06 (3 points) Use the WS 17 reference diagram. For a cup-plus-water boundary, is wall-to-water transfer external or internal? Name one external input and one external output in the diagram.

WS17 - Comparing Heat Transfer Mechanisms

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 4 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS17 reference page preceding these question pages.

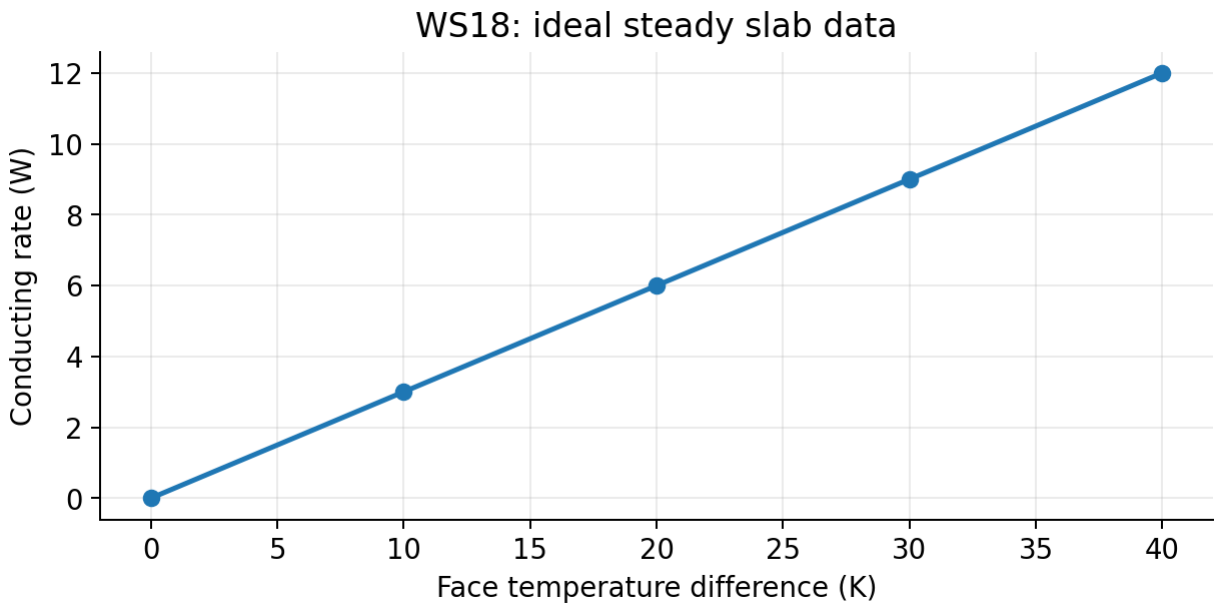
WS17-Q07 (3 points) A system receives 10 W by a separate radiative input and 3 W by conduction, while losing 5 W to air. All rates are constant for 2 min. Calculate net energy input.

WS17-Q08 (3 points) Challenge: A covering reduces a cup’s measured cooling rate. A student concludes it changed only conduction. Explain why that conclusion is too strong and propose a better comparison.

WS18 - Reference page

Keep this page beside the four question pages that follow.

Ideal steady slab values at fixed geometry and uniform conductivity; negligible edge and contact losses.



face_difference_K	rate_W
0	0
10	3
20	6
30	9
40	12

WS18 - Thermal Conductivity

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 1 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS18 reference page preceding these question pages.

Objectives: Use the steady slab model; distinguish material conductivity from whole-object performance; convert area and thickness units; interpret conductivity data.

WS18-Q01 (2 points) State the SI unit of thermal conductivity k and explain why k alone is not a heat-transfer rate.

WS18-Q02 (2 points) Convert 250 cm^2 to m^2 and 8.0 mm to m.

WS18 - Thermal Conductivity

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 2 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS18 reference page preceding these question pages.

WS18-Q03 (2 points) With k,A and face-temperature difference fixed, how does doubling slab thickness affect the ideal steady rate?

WS18-Q04 (2 points) Two equal-geometry slabs have k=.20 and .050 W/(m·K) with the same face temperatures. Find the ratio of their rates, first to second.

WS18 - Thermal Conductivity

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 3 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS18 reference page preceding these question pages.

WS18-Q05 (3 points) A slab has k=.060 W/(m·K), area .40 m², thickness .030 m and face-temperature difference 20 K. Calculate steady conducting rate.

WS18-Q06 (3 points) A steady test conducts 2.4 W through area .020 m² and thickness .010 m with a 15 K face difference. Determine k.

WS18 - Thermal Conductivity

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 4 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS18 reference page preceding these question pages.

WS18-Q07 (3 points) Use the WS 18 reference graph. Find its P-versus-ΔT slope. For A=.10 m^2 and L=.020 m, infer k.

WS18-Q08 (3 points) Challenge: Layer A has k=.040 W/(m·K), thickness .010 m; layer B has k=.060 W/(m·K), thickness .020 m. Compare separate slabs with equal area and equal fixed face differences. Which conducts less, and by what fraction relative to A?

WS19 - Heat Capacity and Specific Heat

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 1 of 4

Difficulty: Beginner through application; final question is a challenge

Objectives: Distinguish C from c and k; calculate signed heat; solve isolated mixing and container-aware calorimetry; check results physically.

WS19-Q01 (2 points) Match k,c,C to these units: W/(m·K), J/(kg·K), J/K. Explain which quantity doubles when a homogeneous sample mass doubles at the same conditions.

WS19-Q02 (2 points) A 250 g sample has c=.80 J/(g·°C). Find its heat capacity.

WS19 – Heat Capacity and Specific Heat

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 2 of 4

Difficulty: Beginner through application; final question is a challenge

WS19-Q03 (2 points) Water cools from 32 °C to 24 °C. Give ΔT and the sign of Q for the water under the sensible-heat model.

WS19-Q04 (2 points) An isolated pair has heat capacities 100 J/K and 300 J/K at 60 °C and 20 °C respectively. Does the equilibrium temperature lie below, at, or above 40 °C? Explain without calculating it exactly.

WS19 - Heat Capacity and Specific Heat

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 3 of 4

Difficulty: Beginner through application; final question is a challenge

WS19-Q05 (3 points) 120 g water with c=4.18 J/(g·°C) warms from 18.0 to 25.0 °C. Calculate heat received, assuming no phase change or work.

WS19-Q06 (3 points) Mix 60 g water at 70 °C with 140 g water at 20 °C in an isolated negligible-capacity cup. Find equilibrium temperature; use the same constant c for both.

WS19 - Heat Capacity and Specific Heat

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 4 of 4

Difficulty: Beginner through application; final question is a challenge

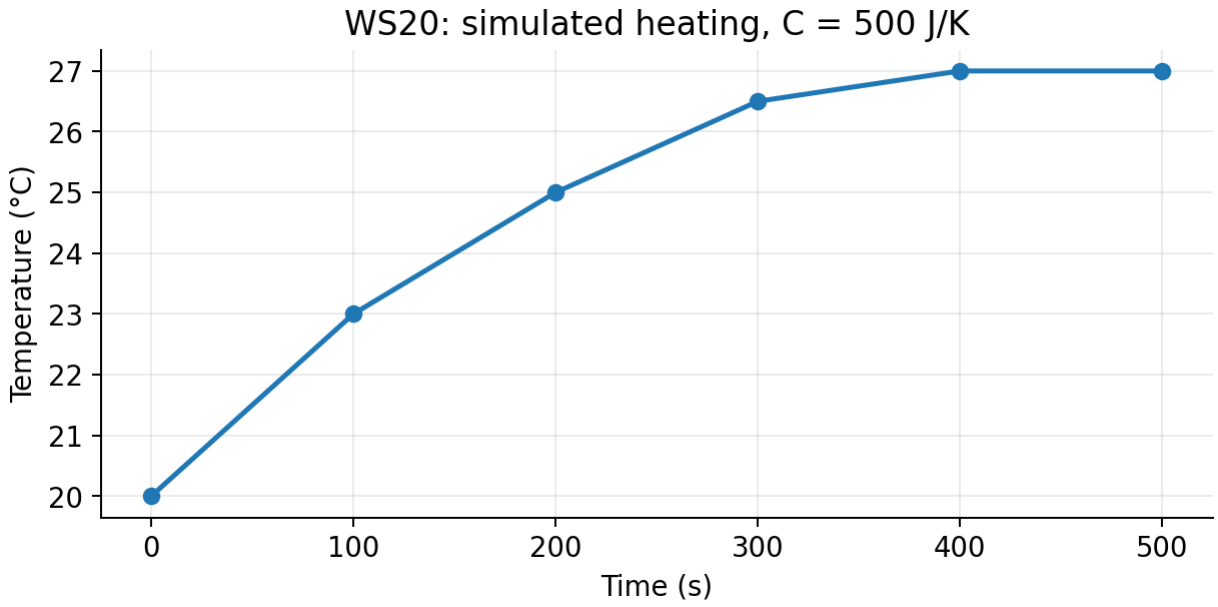
WS19-Q07 (3 points) A 100 g metal at 80 °C is placed in 200 g water at 20 °C. The isolated system reaches 23 °C; cup capacity is negligible. Use water c=4.18 J/(g·°C). Find metal c.

WS19-Q08 (3 points) Challenge: Hot object C=120 J/K at 75 °C; water C=360 J/K and cup C=120 J/K both start at 25 °C. Find final temperature and energy gained separately by water and cup in an isolated model.

WS20 - Reference page

Keep this page beside the four question pages that follow.

Simulated common-temperature cup-and-water system with C=500 J/K, no phase change and negligible work. Absorbed input stays20 W. Use interval averages; these are not actual device measurements.



time_s	temperature_C
0	20
100	23
200	25
300	26.5
400	27
500	27

WS20 – Combined Heat Transfer and Experimental Reasoning

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 1 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS20 reference page preceding these question pages.

Objectives: Combine energy paths with heat capacity; read interval slopes; evaluate controls, uncertainty and device-model limits; justify competition-style predictions.

WS20-Q01 (2 points) A cup-and-water system has absorbed input 10 W and total separate losses 6 W. State net power and whether its temperature rises in a no-phase-change, negligible-work model.

WS20-Q02 (2 points) Why is a lamp’s electrical wattage insufficient to calculate the water temperature rise? Name two missing quantities or processes.

WS20 – Combined Heat Transfer and Experimental Reasoning

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 2 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS20 reference page preceding these question pages.

WS20-Q03 (2 points) A model water-and-cup system has capacity 500 J/K and receives net 2000 J. Find ΔT and final temperature from 21 °C.

WS20-Q04 (2 points) Name two controls for comparing surface treatments using water temperature gain, and explain why final temperature alone can be misleading.

WS20 – Combined Heat Transfer and Experimental Reasoning

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 3 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS20 reference page preceding these question pages.

WS20-Q05 (3 points) 100 g water (c=4.18 J/(g·°C)) and a cup(C=32 J/K) share a temperature. Absorbed input is 14 W; separate losses are 3 W and 2 W. Constant rates act for 250 s from 20 °C. Find final temperature.

WS20-Q06 (3 points) Use the WS 20 simulated curve for C=500 J/K. Calculate average net powers for 0–100 s and 100–200 s. If absorbed input stays 20 W, infer average total losses in each interval.

WS20 – Combined Heat Transfer and Experimental Reasoning

Name: _____ Date: _____

25–30 minutes total | 20 points | Question page 4 of 4

Difficulty: Beginner through application; final question is a challenge

Use the WS20 reference page preceding these question pages.

WS20-Q07 (3 points) Trials A and B show gains 3.0 °C and 3.2 °C. Each initial and final reading has a stated possible error±.1 °C. Give worst-case gain ranges and assess whether the bounds prove B is better.

WS20-Q08 (3 points) Challenge: The WS 20 system levels at 27 °C from 400–500 s while still absorbing 20 W. Explain its net power and loss rate. Why is a prediction made by extending the first 100 s slope to 500 s unjustified?